

Quiz (Carolina Reaper)

Q1. Solve for x in the equation $2^x = 32$.

- (A) $x = 3$
- (B) $x = 4$
- (C) $x = 5$
- (D) $x = 6$
- (E) $x = 7$

Q2. A shipping company can transport 3^x crates per day. If they can transport 243 crates, what is the value of x ?

- (A) $x = 4$
- (B) $x = 5$
- (C) $x = 6$
- (D) $x = 7$
- (E) $x = 8$

Q3. The traffic flow in a city is modeled by the equation $y = 2^{0.5x}$. If $y = 8$, what is the value of x ?

- (A) $x = 2$
- (B) $x = 3$
- (C) $x = 4$
- (D) $x = 5$
- (E) $x = 6$

Q4. Solve for x in the equation $4^x = 16$.

- (A) $x = 1$
- (B) $x = 2$
- (C) $x = 3$
- (D) $x = 4$
- (E) $x = 5$

Q5. A delivery company can deliver x packages per hour, where

$x = 2^y$. If they can deliver 16 packages per hour, what is the value of y ?

- (A) $y = 2$
- (B) $y = 3$
- (C) $y = 4$
- (D) $y = 5$
- (E) $y = 6$

Q6. The time it takes for a package to be delivered is modeled by the equation $t = 3^x$. If $t = 27$, what is the value of x ?

- (A) $x = 1$
- (B) $x = 2$
- (C) $x = 3$
- (D) $x = 4$
- (E) $x = 5$

Q7. Solve for x in the equation $5^x = 125$.

- (A) $x = 2$
- (B) $x = 3$
- (C) $x = 4$
- (D) $x = 5$
- (E) $x = 6$

Q8. A logistics company can store 2^x crates in their warehouse. If they can store 64 crates, what is the value of x ?

- (A) $x = 5$
- (B) $x = 6$
- (C) $x = 7$
- (D) $x = 8$
- (E) $x = 9$

Open Ended Questions (FRQ)

Q9. Solve for x in the equation $x^2 = 2^{2x}$. Show all work and explain your reasoning.

Q10. A transportation company can transport x packages per day, where $x = 3^y$. If they can transport 81 packages per day, what is the value of y ? Show all work and explain your reasoning.

Chef's Tips

- Tip 1: Encourage students to use logarithms to solve exponential equations.
- Tip 2: Remind students to check their work by plugging their answer back into the original equation.
- Tip 3: Emphasize the importance of labeling axes and showing all work when graphing exponential functions.